



Proposed CEA Link — December 2026

A 24-line link built from the December staffing rules and judged on the ORR fatigue factors — nothing carried over from today's roster except the rules

Prepared 8 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules · figures on this page are computed, not typed

PROPOSAL

By the Book

The December rules, built for the fatigue factors

CODE **BB-24-D7**

FINGERPRINT **0f14abce**

BUILT FROM duty table D · seed 7 · 24 lines · 4 cover weeks

LINEAGE Family BB — the workspace's own December duty table (your rules in table form), with the rotation searched for the ORR factors and nothing carried over from today's roster. Compare with "Same Turns" (ST-24-B7 · d15e1b74) and "Five Shift Times" (23 Aug 2026).

✓ **24** lines designed

✓ **35h 00m** a week, exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

✓ **0** fatigue factors present (today: 5)

19 turns

in the December duty table

searched against the timetable; lates shorter than earlies; on the quarter hour except the open and close

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

0 of 25

fatigue factors present

today's link has 5; every other factor is clear or does not apply

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

2h 52m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 — built the other way round from *Same Turns*. Nothing was carried over from today's roster. The duty table is the workspace's own December table: four to open every day, three through to the close and four on a Saturday, five still on at 22:00, fourteen on a Saturday and ten on a Sunday, every duty between 7h and 9h30 on the quarter hour except where it meets the station's open or close, late turns deliberately shorter than early ones, and the times searched against the measured December service. That table pays the 35-hour week to the minute across 20 working lines and four cover weeks.

The rotation was then searched for one thing: the rules it will be assessed against. Chiltern's 13-day limit, twelve hours between duties and the exact contract were treated as walls; the 25 ORR good-practice fatigue factors were the objective, each one scored by the workspace's own `assessFatigue` on every one of about a million candidate arrangements, together with the size of the week-to-week step, weekends off and an even spread of cover weeks. Only after every factor was settled was a week's coherence allowed to count — one turn a week where it costs nothing against the factors — so that a rotation is not chopped up for no fatigue gain. The rules-only version, with no such term, is on page 7 for comparison. Page 7 says how, and how to reproduce it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 5 and 6 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
2	15:15 23:25	16:00 23:55	RD	RD	15:45 23:55	15:45 23:55	16:00 23:55	32h 10m	40h 20m	4
3	14:15 22:25	16:15 23:55	RD	RD	16:15 23:55	16:15 23:55	RD	23h 00m	31h 10m	3
4	RD	14:15 22:00	14:15 22:00	14:15 22:00	14:15 22:00	14:15 22:00	RD	38h 45m	38h 45m	5
5	RD	07:15 16:30	07:15 16:30	07:15 16:30	06:20 15:45	06:20 15:45	RD	46h 35m	46h 35m	5
6	RD	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	16:15 23:55	32h 10m	32h 10m	4
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
8	07:15 16:45	07:15 16:45	RD	RD	06:20 15:30	07:15 16:45	06:20 15:30	37h 20m	46h 50m	4
9	07:15 15:30	06:20 13:30	06:20 13:30	RD	RD	06:20 13:30	07:15 16:45	31h 00m	39h 15m	4
10	08:30 18:00	RD	RD	06:20 13:30	06:20 13:30	06:20 15:30	RD	23h 30m	33h 00m	3
11	RD	15:15 22:45	15:15 22:45	15:15 22:45	15:15 22:45	RD	14:15 22:00	37h 45m	37h 45m	5
12	15:45 23:25	14:15 22:00	14:15 22:00	14:15 22:00	RD	RD	16:40 23:55	30h 30m	38h 10m	4
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
14	16:20 23:25	RD	RD	06:20 14:45	06:20 14:45	06:20 14:45	06:20 13:30	32h 25m	39h 30m	4
15	RD	RD	16:00 23:55	16:00 23:55	16:00 23:55	16:00 23:55	14:15 22:00	39h 25m	39h 25m	5
16	RD	RD	15:15 22:45	15:15 22:45	15:15 22:45	15:15 22:45	15:45 23:55	38h 10m	38h 10m	5
17	RD	RD	16:15 23:55	16:15 23:55	14:15 22:00	14:15 22:00	RD	30h 50m	30h 50m	4
18	RD	06:20 15:45	06:20 15:45	06:20 15:45	RD	RD	08:30 18:00	37h 45m	37h 45m	4
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
20	14:15 22:25	15:15 22:45	RD	RD	07:15 16:30	07:15 16:30	06:20 14:45	34h 25m	42h 35m	4
21	07:15 16:30	06:20 14:45	06:20 14:45	RD	RD	15:15 22:45	15:15 22:45	31h 50m	41h 05m	4
22	RD	RD	07:15 16:45	07:15 16:45	07:15 16:45	07:15 16:45	RD	38h 00m	38h 00m	4
23	RD	07:15 16:45	07:15 16:45	07:15 16:45	07:15 16:45	RD	06:20 15:45	47h 25m	47h 25m	5
24	07:15 14:15	06:20 15:30	06:20 15:30	06:20 15:30	RD	RD	07:15 16:45	37h 00m	44h 00m	4
Cover	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 27m	4.20

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines

How a week compares with today

Nothing here was asked to resemble today's roster; this is how close the fatigue factors took it on their own, and where the December table differs

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	12 of 20
Weeks mixing early and late turns	7	3
Rest-day breaks that are two days or more	13 of 17	22 of 24
Single rest days between duties (not a 48h break)	4	2
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Distinct shift times (none on today's roster)	18	19
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	12	14	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	14	owner's figure for December, leaning late for events
Sunday	8	10	owner's figure for December
Opening at 06:20 (07:15 Sunday)	4	4	unchanged
Through to the 23:55 close, weekday	2	3	three to close
Through to the close, Saturday	2	4	four on a Saturday
Still on duty at 22:00	4	5	five are enough from 22:00

The duty table — the December default, beside today's

DECEMBER TURN	WK	SAT	SUN	TODAY'S TURN (NONE IS REUSED)	WK	SAT	SUN
06:20–13:30 7h 10m · Early	1	1		06:20–13:35 7h 15m	1		
06:20–14:45 8h 25m · Early	1	1		06:20–13:45 7h 25m	1		
06:20–15:30 9h 10m · Early	1	1		06:20–14:00 7h 40m		1	
06:20–15:45 9h 25m · Early	1	1		06:20–14:20 8h 00m	2		
07:15–14:15 7h 00m · Early			1	06:20–14:50 8h 30m		3	
07:15–15:30 8h 15m · Early			1	07:15–15:45 8h 30m			3
07:15–16:30 9h 15m · Early	1		1	08:00–16:30 8h 30m	2		
07:15–16:45 9h 30m · Early	2	2	1	08:30–16:30 8h 00m			1
08:30–18:00 9h 30m · Early		1	1	11:00–19:30 8h 30m	1		
14:15–22:00 7h 45m · Late	2	2		12:00–20:00 8h 00m		1	
14:15–22:25 8h 10m · Late			2	13:00–21:00 8h 00m			1
15:15–22:45 7h 30m · Late	2	1		13:30–21:00 7h 30m		1	
15:15–23:25 8h 10m · Late			1	13:30–22:00 8h 30m	1		
15:45–23:25 7h 40m · Late			1	14:00–22:30 8h 30m	2		
15:45–23:55 8h 10m · Late	1	1		14:30–22:00 7h 30m		2	
16:00–23:55 7h 55m · Late	1	1		14:30–23:25 8h 55m			3
16:15–23:55 7h 40m · Late	1	1		14:45–23:55 9h 10m		2	
16:20–23:25 7h 05m · Late			1	15:15–23:55 8h 40m	2		
16:40–23:55 7h 15m · Late		1					

Read the December table in three blocks: eleven turns Monday to Saturday alike, Saturday's own two (a later 08:30 morning body for its 10:00 peak, and one late spent on a fourth closer), and Sunday's six for its 07:15–23:25 window. Openers stagger by their finish and closers by their start so nobody hands over a cliff; there are only two morning arrival times; and 50 of the 54 start and finish instances sit on :00, :15, :30 or :45 — Saturday's four openers and four closers make a fully quarter-hour day arithmetically impossible.

06:20–13:30 is the one short early; every other early (8h25–9h30) is longer than every late (7h15–8h10), which is the lever against unpopular lates. None of the 19 turns is a time people work today — that is the whole difference between this family and *Same Turns*, and it is the question to put to the room rather than to the tool.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

Monday to Friday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2
PROPOSED		4	7	7	7	7	7	7	7	8	10	10	7	7	7	7	7	5	3
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2
PROPOSED		4	6	7	7	7	7	7	7	8	9	10	8	7	7	7	7	5	4
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3
PROPOSED			4	5	5	5	5	5	5	7	8	8	6	5	5	5	5	5	3
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. Sunday: 5 December movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

Reading it. The table's times were searched against this curve on a five-minute grid, so the cover follows the traffic as closely as the day's fixed minutes allow — the weekday's two peaks are one duty length apart, which is why cover stays fairly flat between them. Saturday's 17:00–22:00 was weighted for events. The proposal raises the whole day rather than reshaping it, which is what the December brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (82.75h) sit on top as RDW, as they do today. Individual weeks range 23h 00m–47h 25m; only the average is the contract.

December design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	✓ 4 weekday · 4 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	✓ 3 weekday · 4 Saturday · 3 Sunday	
Five still on duty at 22:00	✓ 5 weekday · 5 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	✓ 14 · 10	
Four cover weeks, evenly spread	✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6	whole weeks, never scattered days
About 4.2 days a week worked, Mon–Sat	✓ 4.20 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	today's times already obey this
Late turns slightly shorter than most earlies	✓ met — lates 7h15–8h10, earlies 7h10 then 8h25–9h30	one short open turn, then every other early longer than every late

Two things to settle before it is frozen

Familiarity. Every rule is met, and the price is that none of the 19 turns is one people work today — the earlies run to 9h30 and the closers start at 15:45–16:40 rather than 15:15. That is the trade between this family and *Same Turns*, and it is a people question rather than a rules one.

Sunday's finish — five December movements fall after 23:25; the proposal inherits today's window deliberately rather than deciding it.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present · ✓ clear · ● standing, a property of the operation · – not applicable. Nathan assesses against this list.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	– –	– –
FF2	Early shift starting 05:00–07:00 Time of day	● 24	● 24
FF3	Very early shift starting before 05:00 Time of day	✓ 0	✓ 0
FF4	Very early shift (before 05:00) over 8h Duty length	– –	– –
FF5	Day shift over 12h Duty length	✓ 0	✓ 0
FF7	Early shift over 10h Duty length	✓ 0	✓ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	✓ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	✓ 12	✓ 9
FF10	More than 4 consecutive 12h day shifts Cumulative	✓ 0	✓ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	✓ 4
MRSF	More than 12 consecutive day shifts Cumulative	✓ 9	✓ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	✓ 54.4
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	✓ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	– –	– –
FF17	Backward rotating pattern (definition to confirm) Circadian	✓ 10 backward / 12 forward	✓ 10 backward / 12 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	● 20-line rotation · typically 4h 0m a week	● 24-line rotation · typically 2h 52m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	✓ 0
FF6	Night shift over 10h Night shifts	– –	– –
FF8	Less than 2 days rest after consecutive nights Night shifts	– –	– –
FF9	Less than 14h rest in any 24h for night shifts Night shifts	– –	– –
FF12	Only one day rest after night shifts Night shifts	– –	– –
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	– –	– –
FF16	More than 3 consecutive night shifts over 8h Night shifts	– –	– –
FF20	First night shift Night shifts	– –	– –
FF13	Less than 12h rest in any 24h Intervals between duties	✓ 0	✓ 0

Present: 0 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. A cover week's four duties are counted in the worst case for every run figure. This sheet is an aid to a conversation, not a fatigue risk assessment.

How it was chosen, and how to load it

Reproducible: the same table, seed and objective give the same grid on any machine

How it was chosen

1 • The table. The workspace's own December default (links-default-targets.js): the owner's rules — four open, three close (four on Saturday), five on at 22:00, fourteen on a Saturday, ten on a Sunday, four cover weeks — with the times searched against the December demand curve under a strict late-shorter-than-early ordering, fewer distinct times, and the quarter-hour rule. It pays $20 \times 35\text{h}$ exactly; the generator refuses any table that does not.

2 • The grid. Simulated annealing over the 24-line grid: 60,000 candidate arrangements per restart, four restarts from different random starting points per run, four seeded runs — over a million candidates in all — then a steepest-descent polish of every same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

3 • The judge. Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, then every fatigue factor present (each one, and its size: the length of an early block, the hours in seven days, each start-time jump, each backward step), then the week-to-week step and the length of a block on the same sort of week. Then, at a weight no factor can be traded for, the coherence of a week (one turn, its weekend variant) and fairness: four to five days a week, no single rest day between duties. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

Paste it into the workspace

Links → Import, then paste the block on page 8. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied as proposal-import.txt (tab-separated, pastes directly) and proposal.json (the app's own format).

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
BB-24-D7 · Of14abce — By the Book (this proposal)	6	0	6	12/20	2h 52m	41.1	10120
BB-24-D13 · 213b03d8 — runner-up	6	0	5	13/20	3h 32m	41.1	8420
BB-24-D21 · 17dbba5b	6	0	5	11/20	2h 15m	41.1	10550
BB-24-D34 · 9b011899	6	0	5	13/20	3h 23m	41.1	10930
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	41.1	2192436
BB-24-D21p · 5bfe67ed — rules only, no coherence term	6	0	6	1/20	2h 28m	41.1	—
ST-24-B7 · d15e1b74 — Same Turns (the sibling proposal)	6	1	6	18/20	2h 33m	58.5	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	57.7	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December traffic curve (lower is better; today's link scores what it scores). All four candidates clear every factor and share one table, so the fit is identical; full weekends off decided it (6 against 5), and the runner-up is named so it can be asked for. The rules-only row shows what the coherence term cost against the factors: nothing. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

1	SP	SP	SP	SP	SP	SP	SP
2	15:15-23:25	16:00-23:55	RD	RD	15:45-23:55	15:45-23:55	16:00-23:55
3	14:15-22:25	16:15-23:55	RD	RD	16:15-23:55	16:15-23:55	RD
4	RD	14:15-22:00	14:15-22:00	14:15-22:00	14:15-22:00	14:15-22:00	RD
5	RD	07:15-16:30	07:15-16:30	07:15-16:30	06:20-15:45	06:20-15:45	RD
6	RD	15:45-23:55	15:45-23:55	15:45-23:55	RD	RD	16:15-23:55
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-16:45	07:15-16:45	RD	RD	06:20-15:30	07:15-16:45	06:20-15:30
9	07:15-15:30	06:20-13:30	06:20-13:30	RD	RD	06:20-13:30	07:15-16:45
10	08:30-18:00	RD	RD	06:20-13:30	06:20-13:30	06:20-15:30	RD
11	RD	15:15-22:45	15:15-22:45	15:15-22:45	15:15-22:45	RD	14:15-22:00
12	15:45-23:25	14:15-22:00	14:15-22:00	14:15-22:00	RD	RD	16:40-23:55
13	SP	SP	SP	SP	SP	SP	SP
14	16:20-23:25	RD	RD	06:20-14:45	06:20-14:45	06:20-14:45	06:20-13:30
15	RD	RD	16:00-23:55	16:00-23:55	16:00-23:55	16:00-23:55	14:15-22:00
16	RD	RD	15:15-22:45	15:15-22:45	15:15-22:45	15:15-22:45	15:45-23:55
17	RD	RD	16:15-23:55	16:15-23:55	14:15-22:00	14:15-22:00	RD
18	RD	06:20-15:45	06:20-15:45	06:20-15:45	RD	RD	08:30-18:00
19	SP	SP	SP	SP	SP	SP	SP
20	14:15-22:25	15:15-22:45	RD	RD	07:15-16:30	07:15-16:30	06:20-14:45
21	07:15-16:30	06:20-14:45	06:20-14:45	RD	RD	15:15-22:45	15:15-22:45
22	RD	RD	07:15-16:45	07:15-16:45	07:15-16:45	07:15-16:45	RD
23	RD	07:15-16:45	07:15-16:45	07:15-16:45	07:15-16:45	RD	06:20-15:45
24	07:15-14:15	06:20-15:30	06:20-15:30	06:20-15:30	RD	RD	07:15-16:45

Prepared 8 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.